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Tracking the changing patterns of graphic data commentary in economics research articles over time: A local grammar study

Yiming Zhang [a], Lei Zhang [b,*]

[a] Beijing Foreign Studies University, China
[b] Anhui University of Finance & Economics, China

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ABSTRACT

This paper adopts a local grammar approach to explore how the semantic patterns of the discourse act of graphic data commentary in economics research articles have changed over the recent three decades. Based on the comparable corpora of research articles of economics from three time periods (i.e. 1990-1999, 2000–2009, and 2010–2019), we extracted instances of graphic data commentary and performed local grammar analyses to obtain three groups of semantic patterns (i.e. the local grammar patterns). By comparing the frequency distributions of these patterns across the corpora, we find that over the 30 years graphic data commentary has changed to become more concise, interactive, and interpretive. Behind these changes, we argue that the changing social-cultural factors and the disciplinary epistemology of economics may have driven economists to strategically modify the semantic patterns of graphic data commentary. This study also discusses the pedagogical value of the changing patterns of graphic data commentary for current local grammar based EAP teaching.

1. Introduction

Reporting research results effectively plays a pivotal role in academic writing, which can be done either verbally or non-verbally. The non-verbal way, as noted by Swales and Feak (2012), relies heavily on various forms of graphics, including tables, figures, graphs, and charts, to mention only a few. Academic writers not only refer to but often interpret the information in graphics. In this way, graphic results are transferred "from a non-verbal to a verbal mode" (Widdowson, 1979, p. 73), thereby realizing "information transfer" (Swales & Feak, 1995, p. 79). This transfer process involves an important discourse act termed as 'graphic data commentary' (Zhang et al., 2024). In this study, we define graphic data commentary as a discourse act in academic writing to inform and explain the research data in graphics. Previous studies of graphic data commentary tend to be synchronic and largely follow two major approaches: 1) the generic approach (e.g. Swales & Feak, 2012; Wu et al., 2024) which sees graphic data commentary as a task for the rhetorical move of, for example, "summarizing and reporting key results" (Swales & Feak, 2012, p. 368), and 2) the phraseological approach (e.g. Jalilifar et al., 2019; Zhang et al., 2024) which explores the lexical-grammatical or semantic resources of graphic data commentary based on its phraseological behavior.

The aforementioned synchronic studies offer valuable insights into the current state of where graphic data commentary can be performed in academic writing and what language resources can be used to realize this discourse act. However, since the linguistic

* Corresponding author. Administration Building 2, Room 116, English Department, School of Languages and Media, Anhui University of Finance & Economics, 255 Hongye Road, Longzihu District, Bengbu, Anhui, 233041, China.
E-mail addresses: zhangyimingbfsu@163.com (Y. Zhang), zhangleiaufe@126.com (L. Zhang).

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features of academic discourse continuously adapt to changes in communication tools, practices, and sociocultural contexts (Biber & Conrad, 2019), language resources for graphic data commentary are highly likely to change with, for example, the updates of the means of visual representation of research data or the improvement of statistic tools for data processing. To fully investigate graphic data commentary, both synchronic and diachronic studies are needed to reveal this discourse act's status quo and how it has been evolving or developing to its current form. Moreover, examining discourse acts over extended periods can provide a richer context for interpreting the current use of such discourse acts (Baden et al., 2022; Sankoff, 2005).

Researchers in recent years have been aware of the value of diachronic research and there has been an obvious increase in studies concentrating on the diachronic variation of the lexical-grammatical realizations of speech/discourse acts (e.g. Hyland & Jiang, 2018, 2019). Su and his colleagues (i.e. Su, 2020, 2021; Su et al., 2021) drew upon the newly-emerging 'local grammar approach' (Hunston & Sinclair, 2000) in corpus linguistics to perform diachronic investigation of speech/discourse acts. Briefly, local grammar refers to an approach that seeks to describe the lexical-semantic realizations of a particular discourse act or communicative function (Hunston, 2002). This approach was taken as previous studies have shown that local grammar provides a more effective means to analyze and quantify the semantic realizations of quite a few speech/discourse acts. These acts include, for example, evaluation (Hunston & Sinclair, 2000), definition (Barnbrook, 2002), and exemplification (Su & Zhang, 2020). Although Su et al. (2021) have attempted to conduct a local grammar based diachronic study focusing on the academic discourse act of exemplification, the exploration of the value of local grammar in diachronic EAP research is far from systematic. Other discourse acts, especially those more sensitive to the influence of time, such as graphic data commentary (see Kobayashi et al., 2017), merit close attention. Moreover, previous local grammar based diachronic studies have not fully explored the social-cultural and epistemological factors underlying changes in the semantic means for a discourse act. This exploration is essential for enhancing the explanatory power of the findings of local grammar research. In addition, careful thought needs to be given to the way the findings of such studies can be applied to enrich current local grammar based EAP pedagogy (e.g. Zhang & Su, 2021).

To address these issues, this study conducts a local grammar based quasi-diachronic[1] investigation of graphic data commentary in economics research articles, tracking the variation of the semantic patterns of this discourse act over a 30-year time period. It aims to answer the following three questions: 1) Have the semantic patterns for graphic data commentary systematically changed over the recent decades? 2) If yes, what social-cultural and epistemological factors may have triggered the changes? 3) How can the changing patterns of this discourse act be integrated into current local grammar based EAP pedagogy (e.g. Zhang & Su, 2021)?

We chose economics as the subject discipline due to its frequent use of graphical representations to support and establish theories (Demir & Tollison, 2015; Watts & Becker, 2008). More importantly, the graphics in research articles of economics are frequently coupled with a rich discussion and analytical examination of the graphic information (Bhatia, 2010; Maas & Morgan, 2002), which provides a suitable testbed to investigate graphic data commentary. We based our investigation on three self-built corpora of published research articles of economics in three time periods (i.e. 1990-1999, 2000–2009, and 2010–2019), trying to identify the semantic patterns of graphic data commentary in the three corpora from a local grammar perspective and clarify how the semantic patterns of graphic data commentary have developed across the three decades.

2. Literature review

2.1. Research on graphic data commentary

Graphics such as tables and figures have long been recognized as crucial elements in academic writing. For readers, these visual representations highlight important information, reveal relationships and patterns in data, and save reading time, thereby enhancing the accessibility, comprehension, and efficiency of information processing (Franzblau & Chung, 2012). For writers, graphics can complement text descriptions, facilitate succinct communication, and enhance comprehension (Khoury et al., 2019).

For linguists, the discourse act of commenting on graphic data is of high research value. To our knowledge, this discourse act has been most intensively studied in the field of rhetoric research. Early in the 1970s, Widdowson (1978, 1979) noted the interplay between verbal and graphic information and termed it as 'information transfer'. Later, Swales and Feak (1994) borrowed the term and further defined this transfer in academic writing as 'data commentary', referring to the discerning and interpretive verbal commentary that accompanies non-verbal illustrations (p. 83). More importantly, Swales and Feak (2012) observed that data commentary conventionally involves three sub-tasks, pointing out the location of data and summarizing the graphic content, highlighting the information in graphics from the more significant to the less significant, and discussing implications or problems. Each task in turn was found to have some lexical-grammatical features, such as a preference for reporting verbs like *show*, *give*, and *report*, and a consistent use of the present tense as in *Table X shows that*. More recently, Wu et al. (2024) identified the rhetorical functions (e.g. presenting results) and phrase-frames (e.g. *it has been … that*) of graphic data commentary based on a 112,492-word corpus of social science research reports.

Corpus linguists are concerned more with the phraseological features of graphic data commentary (e.g. Jalilifar et al., 2019; Parkinson, 2011; Yang & Wei, 2017). For instance, Parkinson (2011) compared the academic writing of expert and student writers and observed that the two groups differ significantly in their ways of referring to graphics: expert writers tend to provide interpretations and conclusions after referring back to graphics while students prefer making direct claims without mentioning the graphics first.

[1] We use the term 'quasi-diachronic' as this study takes a diachronic perspective to track language changes over a relatively short 30-year period of time. This time span might not be long enough to make this study fully diachronic.

Based on a 7,464,109-word corpus of published research articles in 23 disciplines, Yang and Wei (2017) observed that reporting verbs are frequently used after tables, numbers, ratios, and other inanimate subjects to form a collocational pattern of 'N $_{\text{inanimate}}$ V that'. One key discourse function of this pattern is commenting on graphic data (e.g. *Table 1 shows that*).

These studies have systematically documented the rhetorical and phraseological features of graphic data commentary in academic discourse. However, what specific meanings are communicated to perform graphic data commentary remains unclear. To explore the semantic resources for graphic data commentary, a finer-grained analytical approach is warranted. The study by Zhang et al. (2024) indicates that local grammar can provide an effective means to describe the semantic resources for graphic data commentary.

2.2. Research on local grammar

Local grammar is an analytical framework within corpus linguistics pioneered by Hunston and Sinclair (2000) for the semantic analysis of specific discourse acts or communicative functions. Unlike other grammar approaches (e.g. traditional grammar, construction grammar, and transformational generative grammar) which orient toward the whole of a language, each local grammar targets one language function (e.g. evaluation, exemplification, or definition), revealing its semantic patterns based on analyses of its recurrent word combinations in corpora. A salient feature of local grammar is its use of functionally transparent and context-specific terminologies (e.g. 'Evaluator', 'Evaluation', and 'Evaluative act' in the local grammar of evaluation) rather than conventional grammatical categories (e.g. subject, object, predicate) to analyze the instances of a discourse act. The analytical procedure involves mapping the functional terminologies onto their corresponding syntactic elements within the expressions of a discourse act. Table 1 showcases the local grammar analyses of exemplification (Su & Zhang, 2020).

By matching meanings (e.g. 'Exemplified') with syntactic elements (e.g. *the market economy*) in the instances of exemplification, local grammar describes the semantic realizations of exemplification as a set of semantic patterns, i.e. local grammar patterns like 'Exemplification + Hinge + Indicator + Exemplified' and 'Indicator + Exemplified + Hinge + Exemplification'.

Zhang et al. (2024) analyzed the discourse act of graphic data commentary from a local grammar perspective based on a 982,515-word corpus of published research articles of economics. They identified eight functional terminologies (e.g. 'Data', 'Graphic') from the corpus instances of graphic data commentary and semantically analyzed these instances into 17 frequently occurring local grammar patterns (e.g. 'Graphic + Act + Data', 'Data + Hinge + Graphic'). Their local grammar analyses have been found to be able to effectively capture the subtle semantic variation in the instances of graphic data commentary. For example, local grammar analyses can more precisely reveal the different semantic roles played by the same grammatical functional element of 'Object' in the two instances in Table 2.

As Table 2 shows, in traditional grammar, the two instances both realize an SVO structure. By contrast, in local grammar, they are analyzed as two different patterns as the objects in the two instances realize different meanings/functions: *cumulative net-of-market stock returns* … is analyzed as 'Data' since it only retells the data directly observed from the graphics while *that the relationship exhibits a high degree of consistency* … is analyzed as 'Interpretation' as it provides the academic writer's further understanding or generalization of the graphic data.

Another advantage of local grammar is that the resulting local grammar patterns allow researchers to quantify the semantic means by which a discourse act can be performed. For example, Zhang et al. (2024) quantified the semantic means of graphic data commentary in research articles of economics with 17 local grammar patterns. Each pattern corresponds to a limited number of instances in the corpus. The top local grammar pattern 'Graphic + Act + Data' covers 406 instances of graphic data commentary, with a normalized frequency of 413 times per million words.

The strength of local grammar has also been supported by other studies that analyze different speech/discourse acts such as evaluation (Hunston & Sinclair, 2000) and definition (Barnbrook, 2002; Zhang & Su, 2022), among many others. More recently, some studies have gone even further to explore the potential of local grammars in EAP teaching (e.g. Zhang & Su, 2021; Zhang & Su, 2022). For example, Zhang and Su (2021) proposed a local grammar based EAP teaching method, integrating the local grammar of exemplification into academic writing courses. They found that local grammar patterns have important teaching potential to facilitate EFL learners' performance of discourse acts in academic writing.

2.3. Local grammar based diachronic studies of speech/discourse acts

It is clear that most previous local grammar studies are synchronic, describing the current state of the semantic realizations of speech/discourse acts. In recent years, the 'diachronically oriented discourse analysis' has gained prominence (Brinton, 2015, p. 139) and scholars have developed an intense interest in diachronic pragmatic research of speech/discourse acts. They have conducted qualitative studies based on the observation of a limited number of language instances (e.g. Jucker & Taavitsainen, 2000;

Table 1
Examples of local grammar analyses of exemplification (adapted from Su & Zhang, 2020).

Exemplification	Hinge	Indicator	Exemplified
It	is	an instance of	the market economy.
Indicator	**Exemplified**	**Hinge**	**Exemplification**
An example of	this	would be	her success.

Table 2
Examples of local grammar analyses of graphic data commentary (adapted from Zhang et al., 2024).

Graphic	Act	Data
The chart	shows	cumulative net-of-market stock returns for a sample of companies.
Graphic	**Act**	**Interpretation**
Figure 2	confirms	that the relationship exhibits a high degree of consistency over the 52-year study period.

Pakkala-Weckström, 2008) and quantitative studies based on large corpora of academic English (e.g. Hyland & Jiang, 2017, 2018, 2019).

Against this background, local grammar researchers (Su, 2020, 2021; Su et al., 2021) attempted to conduct diachronic research from a local grammar perspective, investigating changes in the speech act of apology (Su, 2020, 2021) and the discourse act of exemplification (Su et al., 2021) over time. The compatibility of local grammar and diachronic research into speech/discourse acts, as noted by Su (2020), might be due to local grammar's advantage in effectively quantifying the semantic means for a particular speech/discourse act. This quantification enables researchers to more directly observe the changes in the semantic regularities of a speech/discourse act over time based on frequency information instead of close reading of fragmented instances.

Su (2020) made the first attempt at local grammar based diachronic research, focusing on apology in the Corpus of Historical American English (COHA). He retrieved instances of apology in four historical periods (i.e. 1810s–1850s, 1860s–1900s, 1910s–1950s, and 1960s–2000s) from COHA based on lexical items such as *sorry*, *forgive/excuse me*, and *pardon (me)*. Seven terminologies of apology (e.g. 'Apologiser', 'Apology', and 'Forgiveness-seeking') were summarized from the extracted instances. These terminologies were then mapped onto structural elements in the instances to produce 30 local grammar patterns (e.g. 'Apology + Specification', 'Apologiser + Forgiveness-seeking'). Su then compared the distributions of the patterns across the four periods and observed a significant increase in patterns like 'Apologiser + Hinge + Apology', a decrease in patterns like 'Forgiveness-seeking + Apologisee', and a fluctuation of patterns like 'Apologiser + Apology + Specification' over time. In a similar vein, Su and his colleagues further explored the evolution of apology in spoken British English (Su, 2021) and exemplification in research articles of linguistics (Su et al., 2021). These studies further demonstrated that local grammar can serve as a robust means to quantify the semantic patterns of speech/discourse acts and to facilitate the observation of changes in language over time.

Although Su and his colleagues' studies (Su, 2020, 2021; Su et al., 2021) are methodologically significant to have demonstrated the feasibility of applying local grammars in diachronic research, the exploration of local grammar's value in diachronic EAP research remains far from systematic. To date, only one academic discourse act (i.e. exemplification) has been examined diachronically from a local grammar perspective (see Su et al., 2021). Compared with exemplification, the discourse act of graphic data commentary might be more suitable for diachronic research. The reasons are twofold. First, graphic data commentary continuously adapts to innovations in tools for data presentation and analysis (Kobayashi et al., 2017). Second, graphic data commentary involves complicated communicative functions (e.g. generalizing findings, introducing research processes) and intricate form-meaning relationships (e.g. dozens of reporting verbs can be employed to convey the meaning of 'Act') (see Zhang et al., 2024, pp. 77–78), which might also make this discourse act susceptible to the evolution of academic language. Furthermore, previous studies have not adequately addressed the social-cultural and epistemological factors that may drive changes in the semantic means of a discourse act over time, nor have they integrated diachronic findings into the pedagogical applications of local grammar based EAP teaching. To bridge these gaps, the present study takes a local grammar perspective to observe changes in the discourse act of graphic data commentary in economics research articles over the past three decades. Through observation of the changes in local grammar features in the instances of graphic data commentary, our study aims to 1) describe the development of economists' academic language over time, 2) explore the social-cultural and epistemological factors behind the development, and 3) discuss the pedagogical value of the changing patterns of graphic data commentary for teaching discourse acts in EAP settings.

3. Methodology

3.1. Corpus

For the current study, we compiled three corpora of economics research articles, representing the discourse of economics at three points in time, i.e. 1990-1999, 2000–2009, and 2010–2019. Although graphic data commentary is frequently performed in texts from many academic disciplines, the discourse of economics deserves closer attention for two main considerations. Firstly, economics is noted for its heavy reliance on graphics to communicate data and trends (DuPlass, 1996; Watts & Becker, 2008). As DuPlass (1996) highlighted, the discourse of economics is "particularly dependent on the use of the graphic image" (p. 33). Demir and Tollison (2015) also noted that economists rely on graphics to "constitute a theory by themselves and reveal the theory, issue, and significant aspects of the work" (p. 1834). These arguments are supported by Watts and Becker's (2008) survey of 625 college instructors who commonly considered graphics 'important' in econometrics and 'extremely important' in constructing economic theories. Secondly, economics distinguishes itself by rich discussion and analyses centered around graphics (Bhatia, 2010; Maas & Morgan, 2002). As Bhatia (2010) observed, economics is a field deeply engaged in the "discussion and analysis of facts and figures" (p. 41) to enhance understanding of graphic data.

The texts in our corpora were drawn from five prestigious academic journals of economics: *Quarterly Journal of Economics*, *Journal*

of Finance, *Economic Geography*, *Journal of Economic Perspectives*, and *Journal of Economic Growth*. The selection of the five journals was based on three considerations: 1) The journals cover a broad spectrum of topics and sub-disciplines within economics, including macroeconomics, economic development, regional economics, and comparative economic growth. This ensures the representativeness of our corpora. 2) The chosen journals have a long history of publication and have consistently contributed to the development of the field of economics. This makes them suitable for a study that examines the changes in a discourse act over time. 3) All the five journals enjoy high academic influence and status in the field of economics. Specifically, all of them were ranked in the first quartile (Q1) in the impact factor rankings for academic journals of economics in 2021 (based on the Journal Citation Reports released by Clarivate Analytics in 2021).

The corpora we built span three distinct 10-year time intervals on a yearly basis: 1990–1999 (TP1), 2000–2009 (TP2), and 2010–2019 (TP3). Only full-length empirical research articles in the five journals were collected. This could ensure an abundance of instances of graphic data commentary and the consistency of text type across the corpora. As empirical studies rely more heavily on data compared with other studies (e.g. descriptive studies, short papers), they are more likely to be rich in graphics and commentaries to report and discuss graphic data. An equal number of two articles were randomly selected from each journal every year so that each corpus contained 100 texts and all the three corpora were of a similar size (see Table 3 for an overview of the meta-information of the three corpora).

3.2. Working procedure

3.2.1. Data retrieval

We used the concordancer in AntConc to retrieve instances of graphic data commentary in the three corpora. Drawing upon prior research (e.g. Swales & Feak, 2012; Zhang et al., 2024), our primary focus was on typical instances of graphic data commentary with graphics explicitly mentioned (e.g. *Table 1 shows that* …). The instances were retrieved from the corpora by searching for two sets of predetermined lexical items closely associated with graphic data commentary. The first set contained six lexical items listed in Swales and Feak (2012) and Zhang et al. (2024), i.e. *table(s)*, *figure(s)*, *column(s)*, *diagram(s)*, *graph(s)*, and *chart(s)*. It is important to note that commenting on graphic data does not necessarily involve the abovementioned six lexical items (e.g. *it shows that, this indicates that, they reveal that*). To avoid the omission of a considerable number of less explicit instances, we added a second set of lexical items for data retrieval, i.e. 15 reporting verbs for graphic data commentary, as provided in Swales and Feak (2012): *show*, *provide*, *give*, *present*, *summarize*, *illustrate*, *reveal*, *indicate*, *display*, *demonstrate*, *suggest*, *contain*, *depict*, *list*, and *report* (pp. 150–151). Subsequently, we conducted a manual examination of the extracted instances to filter out those which do not perform graphic data commentary. We argue that focusing on these instances would allow us to deal with the most typical expressions of graphic data commentary. We are also confident that searching for these lexical items would enable us to retrieve a considerable number of instances of graphic data commentary for this quasi-diachronic investigation.

3.2.2. Terminology identification

Based on Zhang et al.'s (2024) way of terminology naming, we identified six functional terminologies for graphic data commentary from the retrieved instances (see Table 4). These terminologies are explicit, self-explanatory, and closely associated with the communicative situation of commenting on graphic data. Among these terminologies, 'Data' and 'Interpretation' need further elaboration to clarify their subtle differences. 'Data' differs from 'Interpretation' in that the former involves only retelling the information already shown in the graphics, whereas the latter requires a more careful reading of the graphics so as to provide further understanding of the information in the graphics (as also discussed in Section 2.2).

3.2.3. Local grammar analyses and the quasi-diachronic investigation

Subsequently, we conducted local grammar analyses on the instances by mapping the terminologies in Table 4 onto the structural elements in the instances, as illustrated in Table 5.

Table 5 demonstrates the analyses of three instances extracted from the corpora which produced three different local grammar patterns (see also Zhang et al., 2024 for more detailed instance analyses). Take the instance *These values are shown in Panel B of Figure 10* as an example. The initial noun phrase (i.e. *these values*) which provides direct factual graphic information was termed as 'Data'. The linking verb (i.e. *are*) which functions as 'Hinge' connects the 'Data' with the following reporting verb (i.e. *shown*) realizing 'Act'. The prepositional phrase at the end (i.e. *in Panel B of Figure 10*), which gives the source of data, was analyzed as 'Graphic'. This analysis produced a local grammar pattern 'Data + Hinge + Act + Graphic' which captures the semantic pattern shared by a group of instances of graphic data commentary. Upon the analyses of all the instances from the three corpora, we identified their local grammar patterns and obtained the frequency distributions of these patterns in the corpora. This in turn provided empirical data for the investigation of the changes in graphic data commentary over time. It has to be acknowledged that achieving absolute high accuracy in local grammar

Table 3
Meta-information of the research corpora.

Time period	No. of texts	No. of tokens	No. of sentences
TP1 (1990–1999)	100	882,160	37,786
TP2 (2000–2009)	100	960,296	39,991
TP3 (2010–2019)	100	971,423	40,076

Table 4
Local grammar terminologies of graphic data commentary.

Terminology	Definition	Example
Graphic	Tables, figures, charts, and other visual means to present information	**Figure 1** illustrates this collapse. **In Table 9**, real estate has a positive but small effect …
Act	Actions performed either by graphics or researchers	Table 1 **presents** the descriptive statistics.
Data	The content in graphics, such as specific numbers, trends, or measurements	Figure 1 depicts **these 263 localities and their mortality rates**.
Interpretation	The further understanding or generalization of the information in graphics	Figure 1 suggests **that the secondary loan market was quite resilient during the recent financial crisis**.
Researcher	The person who carries out the act of graphic data commentary	**I** note several additional things about the results summarized in Table 4.
Hinge	Elements that link different functional terms or signal a commentary is being made	This feature **is** also seen in Figure 1. **As is** illustrated in Figure 1, coal prices began to decline from their peak.

Table 5
Illustration of local grammar analyses.

Graphic	Act		Data
Figure 1	depicts		these 263 localities and their mortality rates.
Data	**Hinge**	**Act**	**Graphic**
These values	are	shown	in Panel B of Figure 10
Graphic		**Interpretation**	
From the diagram,		it is clear that the NPG increases as D increases up to a point	

analyses might be rather difficult as the analytical procedure (i.e. form-meaning matching process) involved manual work and subjective choices. However, we are still confident that the analytical results are reliable as 1) our terminologies conform to the two guiding principles of 'Transparency' and 'Economy' for local grammar terminology identification (see also Su, 2020, p. 113), and 2) the accuracy of the local grammar patterns identified is acceptable for the comparison of patterns across time since our analyses were consistent for all the instances in the three corpora.

For data analyses, we relied on the software Log-likelihood and Effect Size Calculator (https://ucrel.lancs.ac.uk/llwizard.html) to conduct log-likelihood ratio analyses on the frequencies of local grammar patterns between each pair of time periods (i.e. TP1-TP2, TP2-TP3, TP1-TP3). The observation of the statistical variation in the local grammar patterns of graphic data commentary over the three time periods can provide a systematic view of the changes in graphic data commentary across the three decades.

4. Results and discussion

Following the analytical procedure demonstrated in Section 3, we analyzed all the instances in the three corpora into three groups of local grammar patterns, which were in turn put into comparative analyses based on their frequency distributions in the corpora. To exclude atypical instances, the following discussion concentrates on the 19 most typical local grammar patterns, each of which has a minimum normalized frequency of ten times per million words in at least one of the three corpora. Those less frequent patterns are detailed in Appendix. Table 6 presents the raw and normalized frequencies of the 19 patterns. To facilitate the observation of the variation in the patterns over time, they were ordered based on their frequency ranking in TP1.

A Chi-Square test for homogeneity revealed a significant overall difference (χ^2 (36) = 548.1, $p < 0.001$) in the distribution of local grammar patterns across the three time periods. To further explore the changes in the use of specific patterns across time, we conducted a series of log-likelihood ratio tests to discern the frequency variation of each local grammar pattern across every pair of time periods (i.e. TP1-TP2, TP2-TP3, and TP1-TP3). As can be seen in the last row of Table 7, the overall frequency of the patterns stayed stable from TP1 to TP2 ($p > 0.05$), which indicated a continuity in the semantic realizations of graphic data commentary from the 1990s to the 2000s. By contrast, the overall frequency increased significantly between TP2 and TP3 ($p < 0.01$) and between TP1 and TP3 ($p < 0.01$). This indicated that economists showed an overall growing trend to perform graphic data commentary over the three decades, with a more notable surge from the 2000s to the 2010s. This preference for graphic data commentary aligned with the substantial advances in graphic processing software and statistical tools since the beginning of the 21st century which have greatly facilitated and deepened researchers' exploration of research data (Li, 2020).

We now turn to a closer observation of the individual patterns to look into the variation in specific semantic patterns of graphic data commentary over time. For clarity, we classified the 19 patterns into four groups based on their frequency trends across the three time periods: 1) increasing patterns (i.e. patterns showing a general growing trend), 2) decreasing patterns (i.e. patterns showing a general declining trend), 3) fluctuating patterns (i.e. patterns showing growing trend first but declining trend later, or vice versa), and 4) stable patterns (i.e. patterns showing no significant growth or decrease over time).

Table 6
An overview of the distributions of the top 19 local grammar patterns across time.

No.	Pattern	TP1		TP2		TP3	
		Raw	Normed	Raw	Normed	Raw	Normed
PA 1[a]	**Graphic** + **Act** + **Data** e.g. Figure 1 depicts these 263 localities and their mortality rates.	527	597	703	732	1016	1046
PA 2	**Interpretation** + **Graphic** e.g. Interviews with foreign subsidiaries confirmed the survey results (Table 5).	269	305	154	160	377	388
PA 3	**Data** + **Hinge** + **Act** + **Graphic** e.g. The results are shown in Table IV.	234	265	181	188	198	204
PA 4	**Graphic** + **Act** + **Interpretation** e.g. Table III shows that this is not true.	192	218	258	269	459	473
PA 5	**Graphic** + **Interpretation** e.g. In Table II, Panel A, our maintained assumption is that crises are exogenous shocks to liquidity and not to the overall structural model of block ownership.	143	162	49	51	127	131
PA 6	**Data** + **Graphic** e.g. The 1989 poverty rate in Montgomery County was 21 percent (Fig. 3).	150	170	220	229	272	280
PA 7	**Data** + **Graphic** + **Act** + **Interpretation** e.g. The results in Table III show striking differences between life and P&C insurers.	94	107	100	104	273	281
PA 8	**Data** + **Graphic** + **Hinge** + **Interpretation** e.g. The results in Table VI are consistent with Hypothesis H2a.	85	96	81	84	148	152
PA 9	**Graphic** + **Researcher** + **Act** + **Data** e.g. In Figure 3 we plot investors' abnormal trading activity in event time.	74	84	99	103	367	378
PA 10	**Researcher** + **Act** + **Data** + **Graphic** e.g. We use the results of the regressions in Table 3.	71	80	110	115	348	358
PA 11	**Hinge** + **Act** + **Graphic** + **Interpretation** e.g. As shown in Table 4, the particular skills they valued were quite different.	41	46	20	21	27	28
PA 12	**Graphic** + **Data** e.g. In columns (1), (2), and (3), the dependent variable is equal to 1.	33	37	57	59	24	25
PA 13	**Interpretation** + **Hinge** + **Act** + **Graphic** e.g. What is really striking is that a constant rate of productivity growth is not an unreasonable assumption for Venezuela, as shown in Figure 6.	29	33	17	18	23	24
PA 14	**Data** + **Graphic** + **Act** + **Data** e.g. The shaded region in Figure IX shows parameters that yield an equilibrium credit limit.	22	25	50	52	68	70
PA 15	**Graphic** + **Hinge** + **Interpretation** e.g. Figure 1 is potentially misleading.	18	20	11	11	13	13
PA 16	**Hinge** + **Act** + **Graphic** + **Data** e.g. As shown in Table 1, DoD-funded IR&D accounts for 7 percent.	10	11	11	11	3	3
PA 17	**Researcher** + **Act** + **Graphic** + **Data** e.g. We also include in Table VI the results of two-sided Tobit regressions.	9	10	25	26	40	41
PA 18	**Data** + **Graphic** + **Hinge** + **Act** e.g. The rate options in Table III were listed.	7	8	14	15	0	0
PA 19	**Hinge** + **Researcher** + **Act** + **Graphic** + **Interpretation** e.g. As we show later (Table IX), changes in discounts are not significantly correlated with interest rate changes.	4	5	9	9	22	23
Total		**2012**	**2281**	**2169**	**2287**	**3805**	**3917**

[a] PA = Pattern.

Table 7
Log likelihood ratios between TP1, TP2, and TP3.

No.	Pattern	TP1-TP2		TP2-TP3		TP1-TP3	
		LL[a]	Sig.	LL	Sig.	LL	Sig.
PA 1	Graphic + Act + Data	12.5	0.000**	94.1	0.000**	114.1	0.000**
PA 2	Interpretation + Graphic	42.2	0.000**	91.9	0.000**	9.2	0.002**
PA 3	Data + Hinge + Act + Graphic	12.0	0.001**	0.6	0.447	7.5	0.006**
PA 4	Graphic + Act + Interpretation	4.9	0.026*[b]	54.8	0.000**	88.6	0.000**
PA 5	Graphic + Interpretation	56.4	0.000**	34.9	0.000**	3.1	0.077
PA 6	Data + Graphic	8.0	0.005**	4.9	0.026*	25.0	0.000**
PA 7	Data + Graphic + Act + Interpretation	0.0	0.873	81.4	0.000**	74.7	0.000**
PA 8	Data + Graphic + Hinge + Interpretation	0.7	0.391	19.1	0.000**	11.7	0.001**
PA 9	Graphic + Researcher + Act + Data	1.8	0.178	160.9	0.000**	185.1	0.000**
PA 10	Researcher + Act + Data + Graphic	5.5	0.019*	127.2	0.000**	173.8	0.000**
PA 11	Hinge + Act + Graphic + Interpretation	1.3	0.002**	1.0	0.325	4.4	0.036*
PA 12	Graphic + Data	4.5	0.032*	14.2	0.000**	2.4	0.119
PA 13	Interpretation + Hinge + Act + Graphic	4.3	0.039*	0.8	0.361	1.39	0.238
PA 14	Data + Graphic + Act + Data	8.9	0.003**	2.6	0.110	20.4	0.000**
PA 15	Graphic + Hinge + Interpretation	2.4	0.125	0.1	0.704	1.4	0.243
PA 16	Hinge + Act + Graphic + Data	0.0	0.981	5.0	0.026*	4.7	0.030*
PA 17	Researcher + Act + Graphic + Data	6.5	0.011*	3.3	0.068	18.3	0.000**
PA 18	Data + Graphic + Hinge + Act	1.8	0.177	19.6	0.000**	10.4	0.001**
PA 19	Hinge + Researcher + Act + Graphic + Interpretation	1.6	0.210	5.5	0.019*	12.0	0.001**
Total		0.1	0.753	435.1	0.000**	402.4	0.000**

[a] LL = Log Likelihood Ratio.
[b] * denotes $p < 0.05$, ** denotes $p < 0.01$.

4.1. Increasing patterns

As can be seen from Table 7, Patterns 1, 4, 6–10, 14, 17, and 19 showed a general trend of growth, covering the largest number of patterns under investigation. Among them, Patterns 1, 4, 6, and 10 were all observed to show a significant increase over all the three period transitions. Four patterns, Patterns 7–9 and Pattern 19, remained stable from TP1 to TP2 but surged significantly from TP2 to TP3, resulting in an overall significant increase from TP1 to TP3. Patterns 14 and 17, on the other hand, showed a significant increase from TP1 to TP2 but remained stable from TP2 to TP3, which also led to an overall significant increase from TP1 to TP3.

A closer reading of the patterns showed that Patterns 1 and 4 are commonly characterized by direct and active graphic information presentation (i.e. introducing a graphic first and then directly showing the data or providing an explanation), which ensures straightforward communication of graphic information. The significant growth of the two patterns over time reflected the changing academic ethos in economics that prioritized pragmatism and efficiency in data presentation. This social-cultural trend, as pointed out by Hyland (2018), was not merely a stylistic preference of a particular discipline but a strategic adaptation to the general evolving demand for clarity and brevity in academic writing.

Patterns 9, 10, 17, and 19 share the meaning 'Researcher' which is often realized by first-person pronouns such as *we* or *I*. This rise of 'self-mention' (Hyland, 2001, p. 207) was consistent with the findings of recent corpus studies (e.g. Hyland & Jiang, 2017; Taylor & Goodall, 2019) which have documented an obvious increase of first-person pronouns in academic writing. To be "personally involved in their text" (Isik-Tas, 2018, p. 34), economists intentionally more frequently used 'Researcher' in the expressions of graphic data commentary. This change in meaning usage might be due to the social demand for international publishing in economics (Mur-Dueñas, 2007), as self-mention can effectively enhance the possibility of international publication by underscoring the originality of research, highlighting researchers' personal contribution, and displaying confidence (Hyland, 2001; Isik-Tas, 2018).

4.2. Decreasing patterns

Patterns 3, 11, 13, 16, and 18 demonstrated a general trend of decline. These patterns are all characterized by the use of passive structures (e.g. *is shown in*, *is illustrated by*). This disfavor of passive structures was in line with Seoane (2013) who also observed a decline in the use of agentless passives since the 1990s. The diminished reliance on passive structures also provided evidence for the trend toward more active and direct commentaries on graphic information over the past 30 years (as discussed in Section 4.1). We argue that this change in semantic patterns might have been driven by the evolving academic writing style in economics that favored transparent and direct meaning communication for quicker access to important scientific data (Millar et al., 2012; Pullum, 2014).

4.3. Fluctuating patterns

Patterns 2, 5, and 12, which fluctuated in frequency over time, belong to Group 3. It is noteworthy that Patterns 2 and 5 share the meaning 'Interpretation' and Pattern 12 is characterized by the meaning 'Data'. The decline of Patterns 2 and 5 and the growth of Pattern 12 from TP1 to TP2 indicated that economists in this period relied less heavily on interpretative analysis of research results and more on the direct presentation of data. It reflected a temporary methodological tendency in economics in the late 1990s in which data-

oriented research characterized by "mathematical forms of explanation" to underscore "economics' connection with quantitative sciences" was highly valued (Demir & Tollison, 2015, p. 3). On the other hand, the significant rise of Patterns 2 and 5, along with the decrease of Pattern 12 from TP2 to TP3, could be interpreted as a resurgence of the value of narrative interpretations in the 2010s, possibly due to their utility in explaining complex economic phenomena and theories (Angrist et al., 2017). The rise of Patterns 7, 8, and 19 (all of which contain the meaning 'Interpretation') from TP2 to TP3 also provided evidence for this trend. As Jockers (2013) noted, researchers increasingly need interpretations to anchor and contextualize quantitative data because rich narrative interpretations could deepen and clarify data analysis and enhance the explanatory power of empirical findings.

4.4. Stable patterns

Only Pattern 15 'Graphic + Hinge + Interpretation' (e.g. *Figure 1 is potentially misleading*) stayed stable across the three time periods, showing no significant growth, decline, or fluctuation in frequency. This pattern is primarily used to make evaluations, which can facilitate the readers' overall understanding of a particular graphic or the content in it. Its stability in frequency over 30 years suggested that evaluation-making has been a constant means by which economists clarify academic concepts and make knowledge.

4.5. Exploration of epistemological reasons

The comparative data of the four groups of patterns showed that most of the patterns underwent obvious changes in terms of frequency of use across the three time periods, showing significant growth, decrease, or fluctuation. The main changes can be summarized as follows: 1) The patterns have become less complicated, focusing progressively on active, clear and effective communication of graphic information. 2) Economists were inclined to present themselves and highlight their personal contributions to the research. 3) Economists showed renewed recognition of the value of qualitative interpretation of research data.

Exploring further, we find that the changing semantic patterns of graphic data commentary reflected the deep epistemological shifts in economics. Firstly, economics has been evolving into "a more pluralistic, less hierarchical, theoretically 'flatter' discipline with greater openness and competition between diverse kinds of approaches within economics" (Davis, 2024, p. 227). This trend was an invitation for researchers from different disciplinary backgrounds to participate in the exploration of research topics in economics, and in turn it demanded simple and clear writing to ensure that the complex economic data and concepts could be easily understood. One of the ways to enhance the readability of academic writing is to provide definite and concise commentaries to make graphics comprehensible. This epistemological need might have caused the changes in the local grammar patterns of graphic data commentary toward semantic simplicity and clarity, clearly seen in the growth of Patterns 1 and 4 and the decline of Patterns 3 and 18 over the three time periods.

Secondly, the trend for more intensive self-mentions in the local grammar patterns was not merely a reflection of publication competition. Epistemologically, it was indicative of the growing awareness of the "conception of self-expression" (Davis, 2024, p. 242) in economics. This awareness may have driven economists to move away from the traditional impersonal writing style toward more visible authorial presence, as evidenced by the rise of Patterns 9, 10, 17, and 19 over the three time periods.

Thirdly, the renewed recognition of the importance of qualitative interpretation (as illustrated in the resurgence of Patterns 2 and 5, the rise of Patterns 7, 8, and 19, and the decline of Pattern 12 from TP2 to TP3) could be related to the growing awareness of 'formal modeling' (Angrist et al., 2017; Hamermesh, 2013) in economics in the recent decade. This epistemological trend required a combination of "numeric, algebraic, and graphical modes of expression" with "verbal statements" (Demir & Tollison, 2015, p. 1834). These verbal statements which take the form of qualitative interpretation can effectively enhance the explanatory power of economic findings.

5. Pedagogical implications for EAP teaching

This study has important implications for EAP teaching, particularly for furthering the local grammar based pedagogy of academic writing (Zhang & Su, 2021; Zhang & Su, 2022), adding a diachronic perspective in teaching the performance of discourse acts.

Drawing on our data on the changing patterns of graphic data commentary over time, we suggest that observed changes in the semantic means for a particular discourse act be integrated into the teaching of this discourse act from a local grammar perspective. This entails both the provision of a systematic inventory of the semantic patterns of a discourse act in the current time period (as attempted in Zhang & Su, 2021) and the explanation of the changing preferences for specific patterns across time periods. The inventory of the patterns provides the basis for discourse act teaching while the explanation can facilitate the learning of the patterns as it provides students with concrete epistemological and social-cultural motivations for the frequent expression of specific meanings by the present-day researchers.

Based on Zhang and Su's (2021) local grammar based academic discourse act teaching method, this study suggests that explanation of variation in local grammar patterns over time be added into Zhang and Su's four-step teaching procedure. Specifically, in Zhang and Su's (2021) step of "identifying functional terminologies for analyzing the discourse act" (p. 3), we suggest a fuller elucidation of certain functional terminologies based on their prominence in certain time periods. For example, when introducing the terminology of 'Researcher' for graphic data commentary in the academic writing of economics, the teacher can 1) highlight that economists have increasingly acknowledged the importance of the meaning 'Researcher' over the last three decades, and 2) further explain that the presence of 'Researcher' agrees with the epistemological trend for stronger conception of self-expression in the academic writing of economics.

Table 8
Variation of the local grammar patterns containing 'Researcher' over time.

Pattern	Frequency			
	1990–1999 (normed)	2000–2009 (normed)	2010–2019 (normed)	Trend
Pattern 9 Graphic + **Researcher** + Act + Data	84	103	378	↑
Pattern 10 **Researcher** + Act + Data + Graphic	80	115	358	↑
Pattern 17 **Researcher** + Act + Graphic + Data	10	26	41	↑
Pattern 19 Hinge + **Researcher** + Act + Graphic + Interpretation	5	9	23	↑

To validate the explanation, in the subsequent step of "identifying local grammar patterns" (Zhang & Su, 2021, p. 5), the teacher shows corpus instances of the local grammar patterns containing the terminologies under explanation. One key procedure is to demonstrate the frequency variation of the local grammar patterns across time. For example, to illustrate the variation of the meaning 'Researcher', Patterns 9, 10, 17, and 19, which share this meaning and showed significant frequency variation across time, can be collected from Tables 6 and 7 to help students clearly see the upsurge of this meaning over the last three decades (as shown in Table 8).

To further strengthen students' awareness of changes in the semantic patterns of a discourse act over time, we suggest adding a new classroom exercise to Zhang and Su's (2021) step of "practicing contextual use of local grammar patterns" (p. 5). This exercise involves a rewriting task in which students are given, for example, instances of graphic data commentary more typical in the 1990s and are then required to rewrite the instances following the local grammar patterns more typical in the 2010s. For example, an instance from TP1 is presented first (see Example 1).

(1) The changing trade intensities are shown in Table 1.
(2) In Table 1, we show the changing trade intensities.

Students are then required to rewrite Example 1 following the local grammar pattern 'Graphic + Researcher + Act + Data'. When the sentence has been successfully rewritten (see Example 2), students are reminded that the original sentence characterized by the absence of the author (Example 1) was popular in the 1990s and it has developed into the more recent interactive expression (Example 2). By comparing the use of the patterns by present-day researchers with their colleagues in the past, students can be better informed about the acceptability of certain semantic patterns for contemporary research. The evolving disciplinary epistemology is then re-explained to justify the changing semantic patterns of graphic data commentary, which allows students to better understand why current expressions are as they are and how to follow the epistemological trends of a particular discipline with semantic and lexical resources.

6. Conclusions

This study has conducted a local grammar based quasi-diachronic investigation of graphic data commentary in research articles of economics, probing into the systematic changes in the semantic patterns of this discourse act from the 1990s to the 2010s. Our analyses revealed that when commenting on graphic data, economists tended to 1) use more concise and active expressions, 2) engage in the discourse, and 3) add more interpretations across the 30 years. These tendencies were in line with the broader social-cultural and epistemological changes in the discipline of economics: 1) growing recognition of brevity and clarity in academic language to follow the epistemological trend for higher readability of academic writing, 2) the trend for more intensive self-expression (Davis, 2024) to enhance the originality and interactivity of research in publication competition, and 3) the renewed recognition of qualitative interpretation which echoed the epistemological trend for 'formal modeling' (Angrist et al., 2017), i.e. more in-depth data discussion in economics.

These findings further indicate that local grammar is methodologically applicable for diachronic EAP research. Through transparent description and systematic quantification of the semantic patterns for a particular academic discourse act, local grammar is in a better position to 1) capture the subtle meanings that characterize a specific time period, 2) clarify how the semantic means for a discourse act can systematically change across time, and 3) explore deep into the social-cultural and epistemological factors behind such semantic changes in the discourse act. Pedagogically, this study can enrich current local grammar based academic writing teaching method (e.g. Zhang & Su, 2021) by integrating a diachronic perspective into the teaching of the semantic patterns of academic discourse acts.

This study also has limitations. First, we are still in need of a more robust means to identify instances of graphic data commentary. Although this study employed two sets of predetermined lexical items to retrieve language instances, some instances which lack explicit lexical markers still escaped our retrieval process. Second, while our observation of the three-time-period corpora has revealed important tendencies in graphic data commentary variation, the time span can be further extended for a standard diachronic study so that the evolution of this discourse act can be more comprehensively observed. These issues deserve further exploration in future studies.

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Appendix. Less frequent patterns and their distributions in the corpora

No.	Pattern	TP1		TP2		TP3	
		Raw	Normed	Raw	Normed	Raw	Normed
PA 20	**Researcher** + **Act** + **Interpretation** + **Graphic** e.g. We show that the positive relationship between inequality and differential fertility is primarily found in the sample consisting of middle-income countries, while among high income countries the relationship is weaker, and among the low-income countries the relationship is weakest (Table 4).	0	0	7	7	4	4
PA 21	**Data** + **Hinge** + **Graphic** e.g. The data are in Table IV.	0	0	3	3	0	0
PA 22	**Interpretation** + **Hinge** + **Graphic** + **Act** e.g. They are also robust to the inclusion or not of the investment share in the regression, as Table 3 shows.	0	0	2	2	3	3
PA 23	**Data** + **Researcher** + **Act** + **Graphic** + **Interpretation** e.g. The various models we report in Table III generally lead to a similar impact of the increase in cash flow volatility on the cash ratio, except for the changes model and the fixed effects model, where the effect is also positive but lower.	0	0	2	2	0	0
PA 24	**Interpretation** + **Hinge** + **Researcher** + **Act** + **Graphic** e.g. These traditional expected return estimates seem too volatile to be plausible, as we also show in Figure 4.	0	0	2	2	1	1
PA 25	**Data** + **Researcher** + **Act** + **Interpretation** + **Graphic** e.g. The second dependent variable we report relates closely to weekly earnings for salaried workers in the retail industry (Table 2).	0	0	1	1	0	0
PA 26	**Data** + **Act** + **Graphic** + **Interpretation** e.g. The percentiles presented in Table 6 offer more information on the distribution of the effects.	0	0	1	1	0	0
PA 27	**Researcher** + **Act** + **Graphic** e.g. We demonstrate Tables VI and III.	0	0	1	1	0	0

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Yiming Zhang is currently a postgraduate of English linguistics and applied linguistics at Beijing Foreign Studies University, China. His research interests are pragmatics and corpus linguistics. He has published several articles in Chinese journals. Email: zhangyimingbfsu@163.com

Lei Zhang received his PhD in Beihang University, China and is currently an associate professor at Anhui University of Finance and Economics, China. His research interests mainly are corpus linguistics and English language teaching. He has published articles in *English for Specific Purposes* and *Journal of English for Academic Purposes*, and several articles in Chinese journals. **Email:** zhangleiaufe@126.com